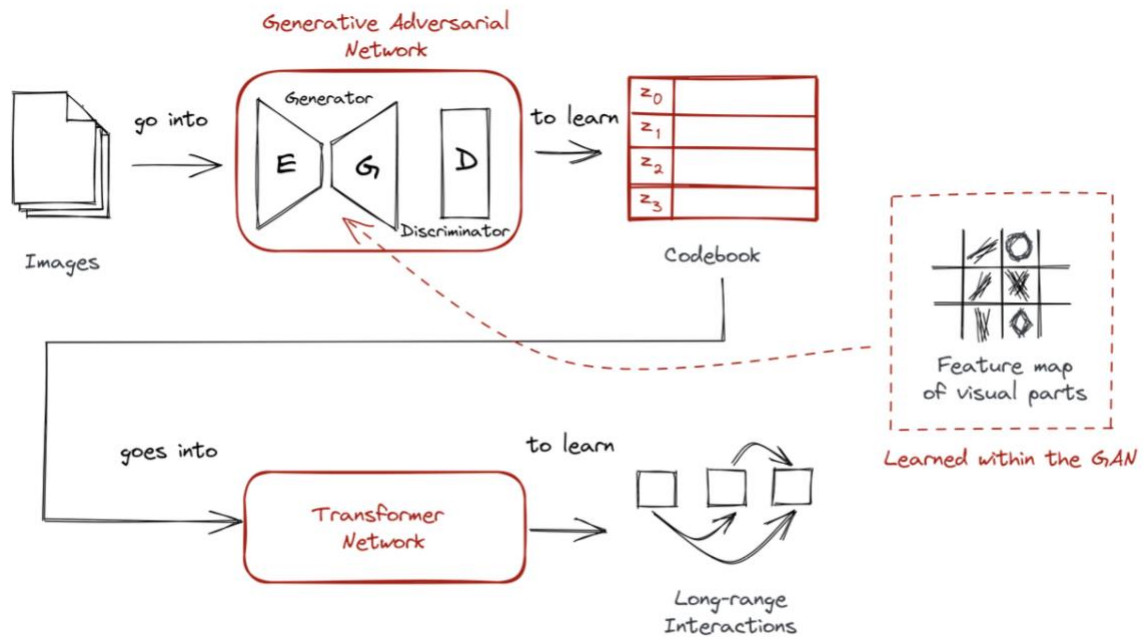
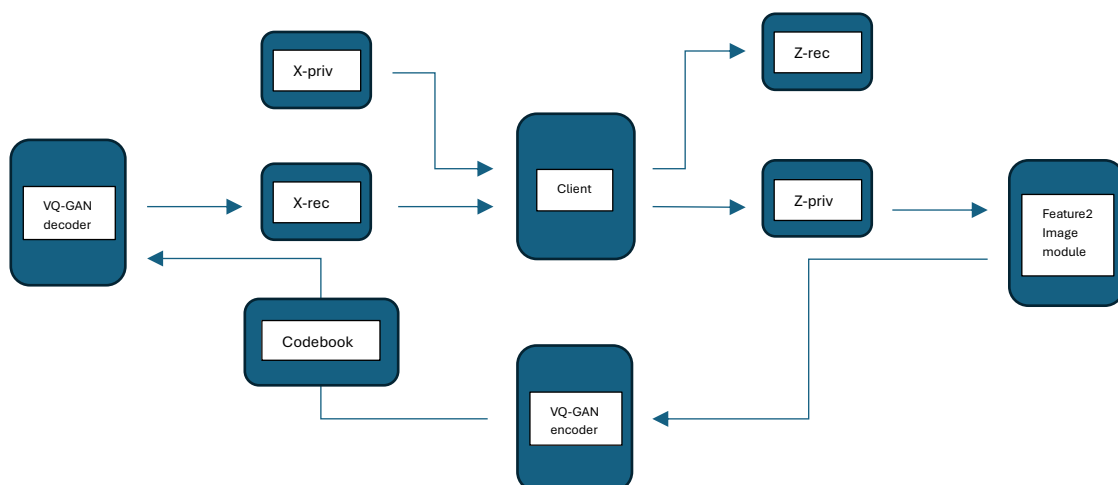


Attempting to replace Style-GAN with VQ-GAN in GAN Based Latent Space Search for Data Reconstruction Attack.

VQ-GAN architecture:



Proposed Framework (rough figure):



VQ-GAN based latent space search takes in the client's intermediate feature representation (Z-private) and tries to reconstruct the private inference instance (X-private). In the original paper they used a pretrained style-GAN, we would also like to use a pretrained VQ-GAN. This framework consists of an encoder, codebook and a decoder. The decoder will use the updated codebook indices to reconstruct X-private, which is X-reconstructed. Here X-rec and X-priv are passed through the client model to generate the intermediate feature representation that Z-rec and Z-priv respectively. We will then compare the Z-rec and Z-priv using the loss function to optimize the codebook indices. The Z-rec is then feeded into the feature2image model to convert it into an image. This image is then used in the VQ-GAN encoder to generate new codebook indices. This process is repeated until we achieve success in reconstructing X-rec that is as close as it can be to the private inference X-priv .

Challenges:

1. The main challenge would be the change from continuous latent space to discrete latent space, as unlike Style-GAN, VQ-GAN uses a codebook that is used by the transformers.
2. The Loss function will now include perceptual loss (like reconstruction loss) and commitment loss.
3. Using a different method for latent space search handling and optimization as we are working with discrete latent space z_q .